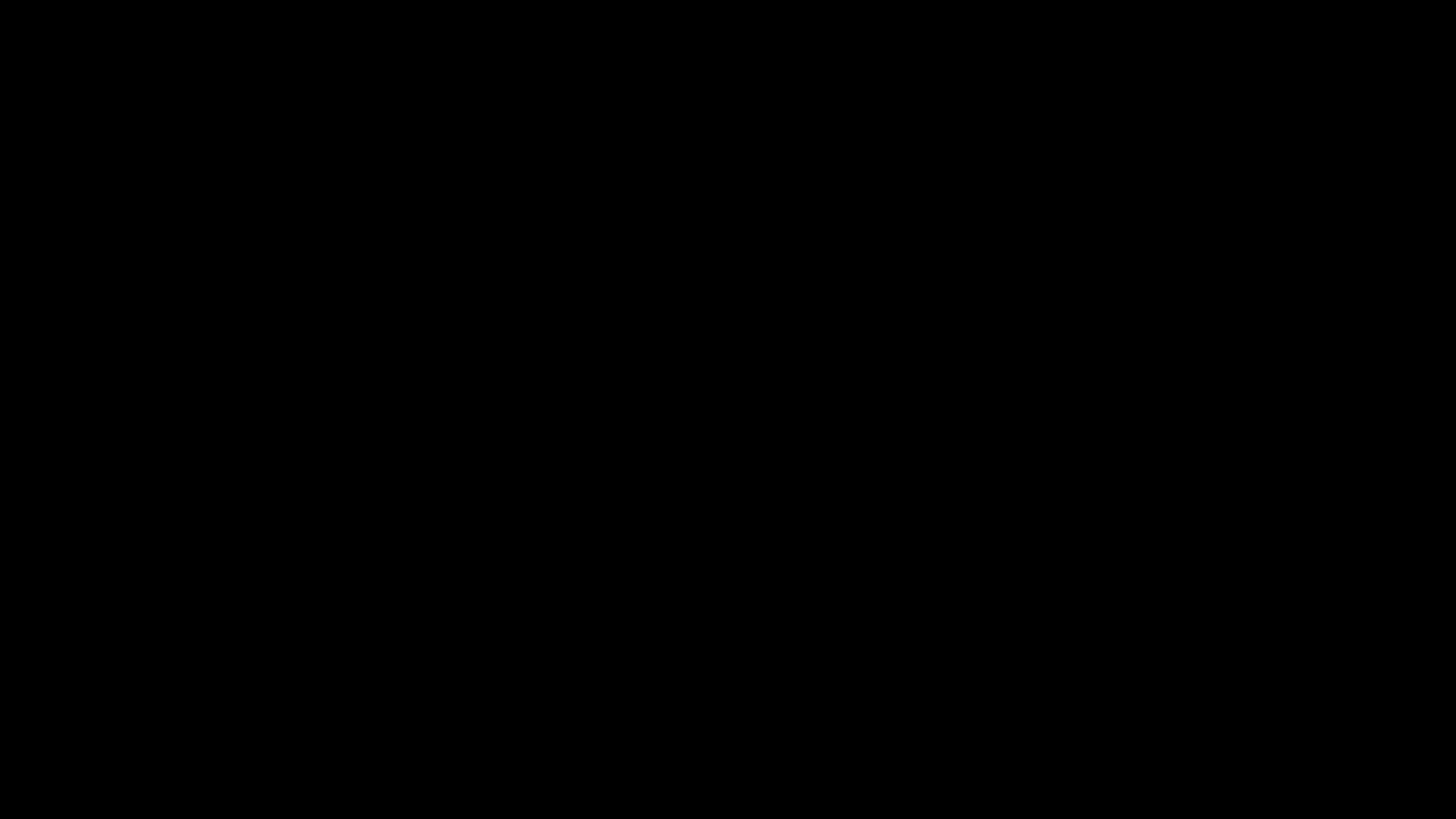




- 1) Pace ↑
- 2) No Backlogs

Strings

In Java





Contents

1. String Basics
2. Built-In Methods
3. Immutability
4. StringBuilder
5. Questions

character Array

```
char[] arr = {'r', 'a', 'g', 'h', 'a', 'v'};
```

```
cout(arr[3]);
```

strings

String as a data type



String Data Type

charAt() & length()

0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19
 String s = "Chaitanya and Shruti";

sout(s.charAt(4)); // s[4]

int n = s.length();

Ques: Count the number of vowels in a given string

'a', 'e', 'i', 'o', 'u'

Ques: Palindrome String

mom

$O(n)$

dad

racecar

malayalam

racecar
i j

naman

madam

nitin

abccba

indexOf() & compareTo()

s1.compareTo(s2);

↓
integer

lexographically smaller

raghav < sneha

harmeet < harsh

harsh harshit

Raghav < aditya

`contains()` & `startsWith()`

Method	Parameter	Return Value
<code>contains()</code>	<code>String</code>	<code>boolean</code>
<code>startsWith()</code>	<code>String</code>	<code>boolean</code>

toLowerCase() & concat()



```
s1.concat(s2);
```



Ques: Change the string

String + int/char/String



sout("raghav"+10+20)

left to right

raghav30
α

raghav1020 ✓

sout (10 + 20 + "raghav");

sout (10 + "raghav" + 20);



Ques: Take integer input and convert it into a string



Ques: Digits in a given number

substring(i) & substring(i,j)

Ques: Print all substrings

Array to String

$s = \text{"gopi"}$

$g, go, gop, gopi$

o, op, opi

p, pi

i

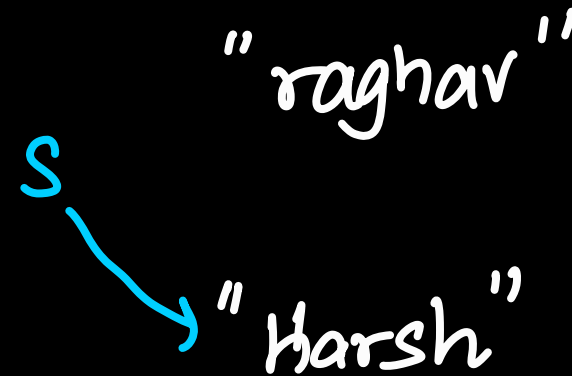


HW: Sum of all substrings of a number

Interning & new keyword

```
String s = "raghav";
```

```
s = "Harsh";
```

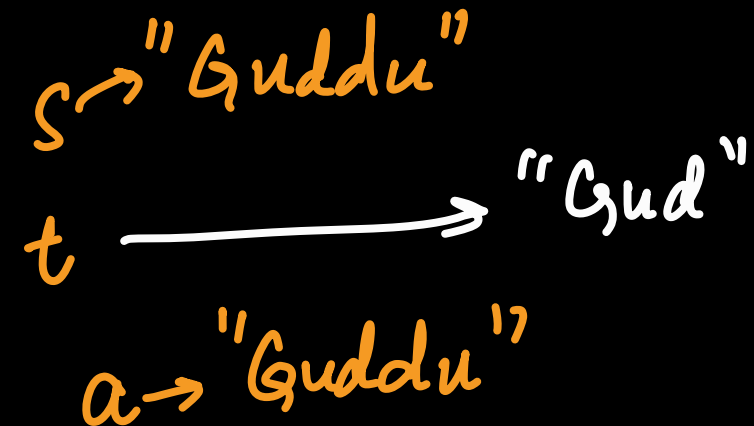


```
String s = "Guddu";
```

```
String t = "Guddu";
```

```
t = "Gud"
```

```
String a = new String("Guddu");
```



Interning & new keyword

```
String s = "Karan";
```

```
s += "Saini";
```



```
t = "";
```

```
t += 'r';
```

```
t += 'a';
```

```
t += 'g';
```



String immutability in Java



cannot change individual characters

Security purpose

0 1 2 3 4 5
s = "Bishal";
 ↓
 m

s = s.substring(0,3) + 'm' + s.substring(4);

s1.equals(s2)

```
String s1 = "Raghav";
```

```
String s2 = new String("Raghav");
```

```
cout ( s1 == s2 ) ;
```

↓
false

StringBuilder



setCharAt()



Ques: Toggle Characters

s1.append(s2)



insert() & deleteCharAt()

Method	Parameter	Return Value
insert()	int index, String str	String
deleteCharAt()	int index	String

reverse()



Ques: Reverse each word in a given string

$S = \text{"Bishal is a bad boy"};$
 $\downarrow$
 $ans = \text{"lahsib si a dab yob"};$

$S = \text{"Archit is star"};$
 $\downarrow$
 $ans = \text{"tihcra si rats"};$

Ques: Anagram

fired

fried

latent

talent

anagram

nagaram

race

care

listen

silent

acer

acer

l-1

i-1

s-1

t-1

e-1

n-1

Ques: Most frequent character

String s = "character";

c-2

a-2

r-2

Method-1 : Brute Force

T.C. = $O(n^2)$ "Nested Loop"

Method-2 : Sorting T.C. = $O(n \log n)$

s = "aacc**e**h**r**rt";
 i
 j



Sliding Window

Ques: Most frequent character

$s = "aaccettzzzz"$

i
j

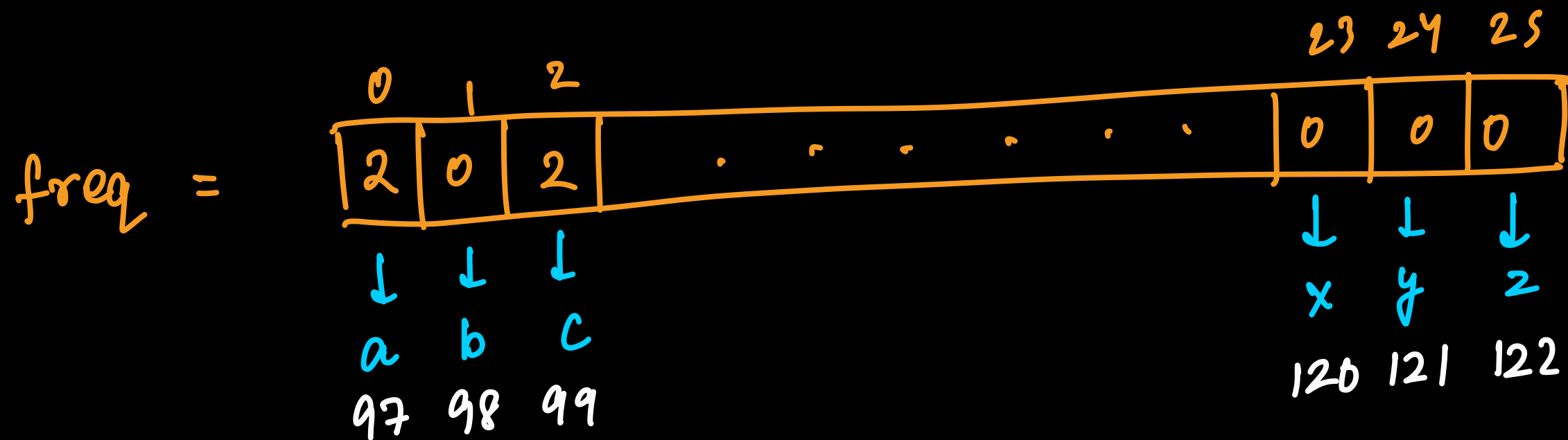
maxFreq = 1 2 3 4

ans = a ~~t~~ z

Ques: Most frequent character

Method-3 : Frequency array.

$S =$ character



A.S. = $O(1)$ T.C. = $O(n)$



HW: Non repeating character



Ques: String Compression [leetcode]

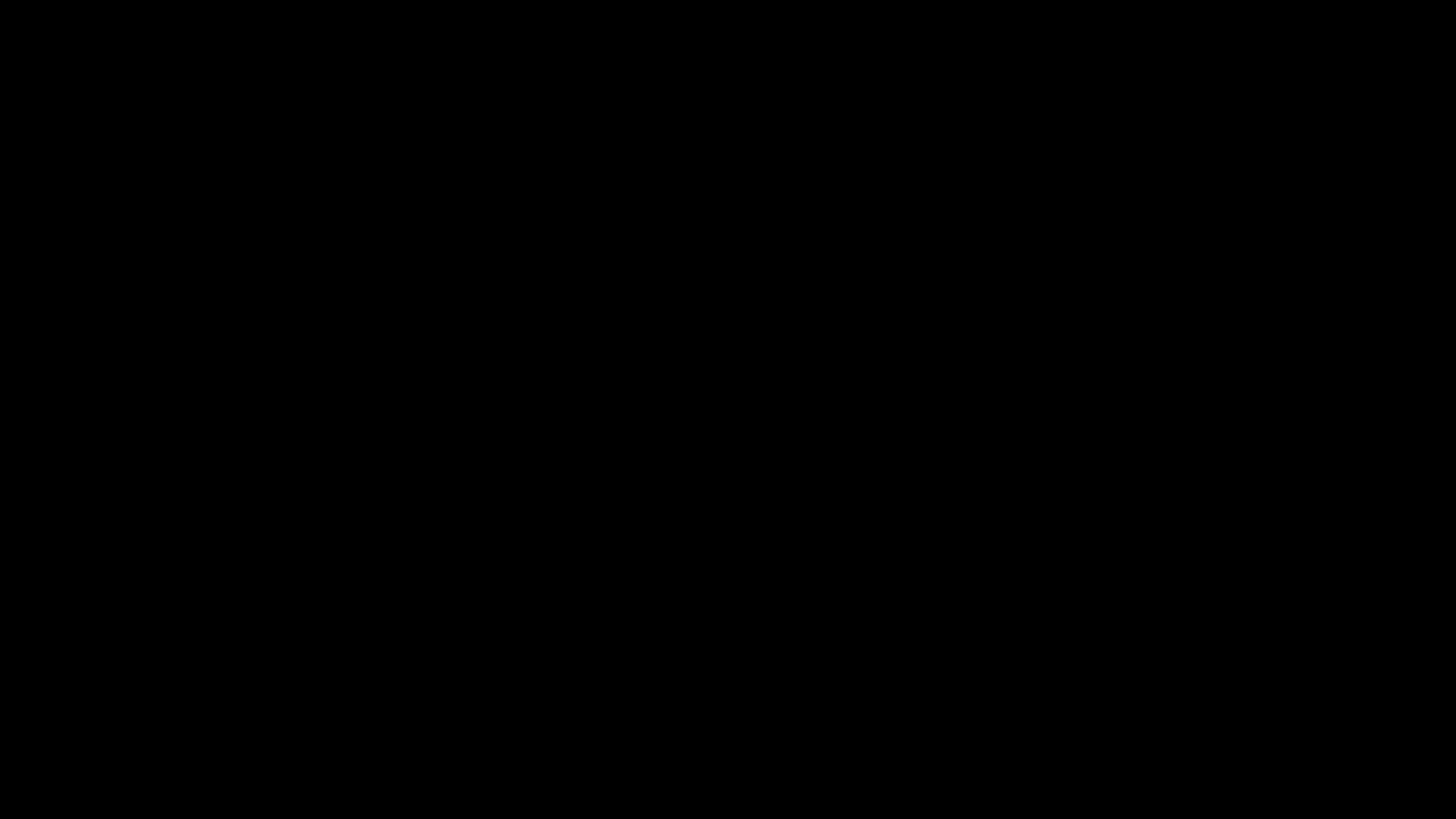
$s = "a^i a a b b z z t k k a a"$



$ans = "a3b2z2tk2a2"$



HW: Easy String





THANKYOU

Cuties